

2016 EKAINA A4

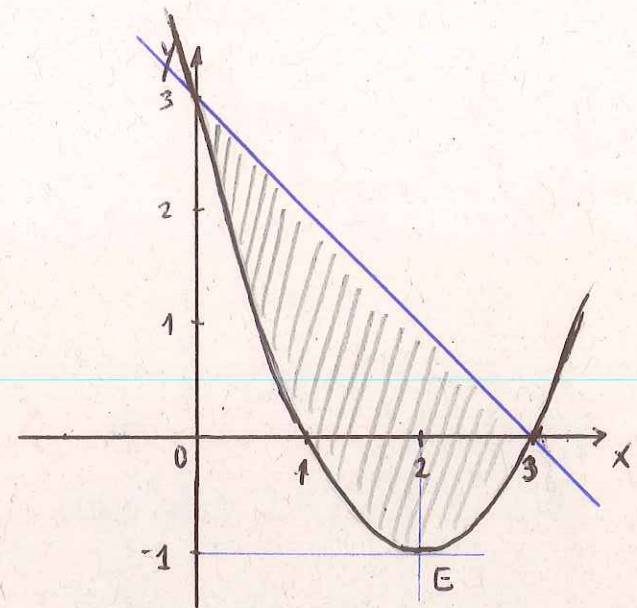
x ardatzarekiko ebaki-puntuak

$$f(x) = x^2 - 4x + 3 \rightarrow x^2 - 4x + 3 = 0 \begin{cases} x=1 \rightarrow (1,0) \\ x=3 \rightarrow (3,0) \end{cases}$$

$$g(x) = -x + 3 \rightarrow (0,3) \text{ eta } (3,0) \text{ puntuetatik pasatzen da adibidez.}$$

Parabola: $f(x) = x^2 - 4x + 3$

erpina: $x = \frac{-b}{2a} = \frac{4}{2} = 2 \Rightarrow E(2, -1)$



1. Ebaki-puntuak funtzioen artean

$$x^2 - 4x + 3 = -x + 3$$

$$x^2 - 3x = 0$$

$$x(x-3) = 0 \begin{cases} x=0 \rightarrow (0,3) \\ x=3 \rightarrow (3,0) \end{cases}$$

2. Kenketa funtzioa

$$y = (-x+3) - (x^2-4x+3) \rightarrow \boxed{y = -x^2 + 3x}$$

3. Integrala

$$\int_0^3 (-x^2 + 3x) dx = \left[-\frac{x^3}{3} + \frac{3x^2}{2} \right]_0^3 = \left(-\frac{3^3}{3} + \frac{3 \cdot 3^2}{2} \right) - (0 - 0) = \frac{9}{2}$$

4. Azalera

$$\boxed{A = \frac{9}{2} u^2}$$

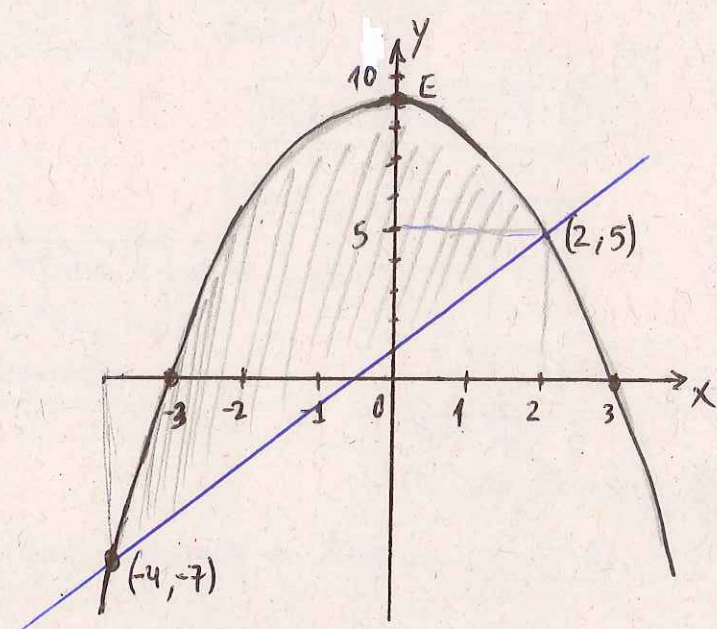
2016 UZTAILA B4

$$y = 9 - x^2$$

$$y = 2x + 1 \rightarrow (1,3) \text{ eta } (0,1) \text{ puntuetatik pasatzen da adibidez}$$

Parabola: $y = 9 - x^2 \rightarrow$ erpina $x=0$
 $E(0,9)$

x ardatzarekiko ebaki-puntuak $\rightarrow 9 - x^2 = 0$
 $x = \pm 3$



1. Ebaki-puntuak:

$$9 - x^2 = 2x + 1 \rightarrow -x^2 - 2x + 8 = 0$$

$$x \begin{cases} x = -4 \rightarrow (-4, -7) \\ x = 2 \rightarrow (2, 5) \end{cases}$$

2. Kenketa funtzioa

$$y = (9 - x^2) - (2x + 1) \rightarrow y = -x^2 - 2x + 8$$

3. Integrala

$$\int_{-4}^2 (-x^2 - 2x + 8) dx = \left[-\frac{x^3}{3} - x^2 + 8x \right]_{-4}^2 = \left(-\frac{8}{3} - 4 + 16 \right) - \left(\frac{4^3}{3} - 16 - 32 \right) = 36$$

4. Azalera

$$\boxed{A = 36 u^2}$$

2015 EKAINA A4

$$f(x) = x^2 - 2x + 1$$

$$g(x) = -x^2 + 5$$

f(x) parabola

$$\text{erpina} \rightarrow x = \frac{-b}{2a} = \frac{2}{2} = 1 \rightarrow E_1(1, 0)$$

X ardatzarekin ebaki-puntuak:

$$x^2 - 2x + 1 = 0 \rightarrow \begin{cases} x = 1 \\ x = 1 \end{cases} \rightarrow \text{erpina da}$$

g(x) parabola

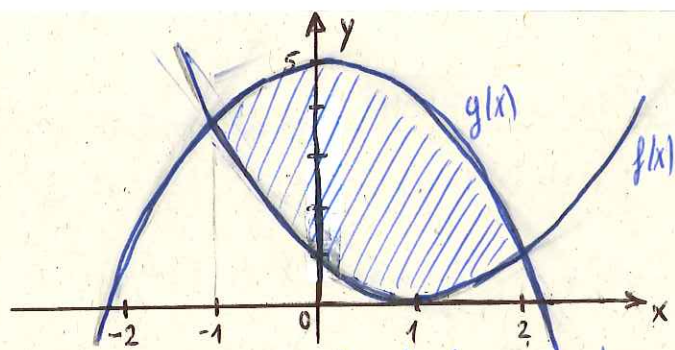
$$\text{erpina} \rightarrow x = \frac{-b}{2a} = 0 \rightarrow E_2(0, 5)$$

X ardatzarekin ebaki-puntuak

$$-x^2 + 5 = 0 \rightarrow \begin{cases} x = -\sqrt{5} \\ x = \sqrt{5} \end{cases}$$

$$= \left(\frac{-2^4}{3} + 4 + 8 \right) - \left(\frac{2}{3} + 1 - 4 \right) = 9$$

4. Azalera $A = 9u^2$



1. Ebaki-puntuak funtzioen artean

$$x^2 - 2x + 1 = -x^2 + 5$$

$$2x^2 - 2x - 4 = 0 \rightarrow \begin{cases} x = -1 \rightarrow (-1, 4) \\ x = 2 \rightarrow (2, 1) \end{cases}$$

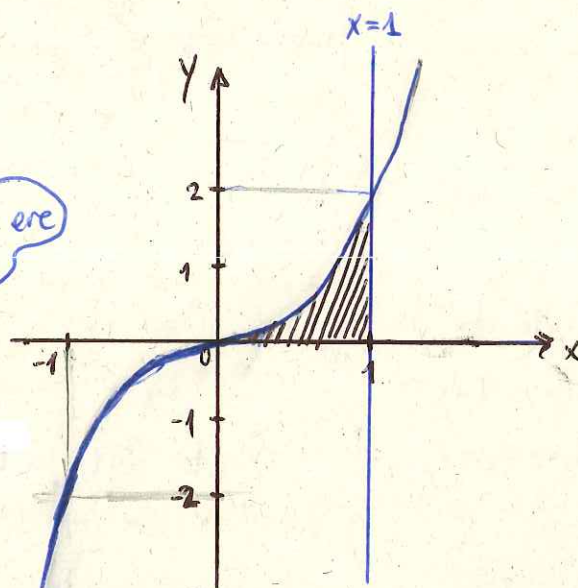
2. Kenketa funtzioa

$$y = (-x^2 + 5) - (x^2 - 2x + 1)$$

$$y = -2x^2 + 2x + 4$$

3. Integrala

$$\int_{-1}^2 (-2x^2 - 2x + 1) dx = \left[\frac{-2x^3}{3} - x^2 + x \right]_{-1}^2 =$$



2015 UZTAILA B4

$$y = 2x^3 \rightarrow y' = 6x^2$$

$$x = 1$$

$$y = 2x^3$$

$$(-1, -2) \quad (0, 0) \quad (1, 2)$$

$$y'(0) = 0 \rightarrow \begin{cases} X ardatza da \\ (0, 0) \text{ puntuko} \\ \text{ukitzailea} \end{cases}$$

Irudian ere ikusten da

1. Ebaki-puntuak $\rightarrow$ zuzenean grafikotik $\begin{pmatrix} x=0 \\ x=1 \end{pmatrix}$

2. Kenketa-funtzioa $\rightarrow$ Ez dago inolakorik $\Rightarrow y = 2x^3$ integratu behar da bakarrik

3. Integrala $\int_0^1 2x^3 dx = \left[\frac{x^4}{2} \right]_0^1 = \frac{1}{2} - 0 = \frac{1}{2}$

4. Azalera $A = \frac{1}{2}u^2$

2014 EKAINA A4

X ardatzarekiko ebaki-puntuak

$$y = -x^2 + 2x \rightarrow \text{erpina } x = \frac{-2}{-2} = 1 \rightarrow E_1(1, 1)$$

$$y = x^2 - 10x \rightarrow \text{erpina } x = \frac{10}{2} = 5 \rightarrow E_2(5, -25)$$

$$-x^2 + 2x = 0 \rightarrow x(-x + 2) = 0 \begin{cases} x=0 \\ x=2 \end{cases}$$

$$x^2 - 10x = 0 \rightarrow x(x - 10) = 0 \begin{cases} x=0 \\ x=10 \end{cases}$$

1. Ebaki-puntuak

$$-x^2 + 2x = x^2 - 10x$$

$$-2x^2 + 12x = 0$$

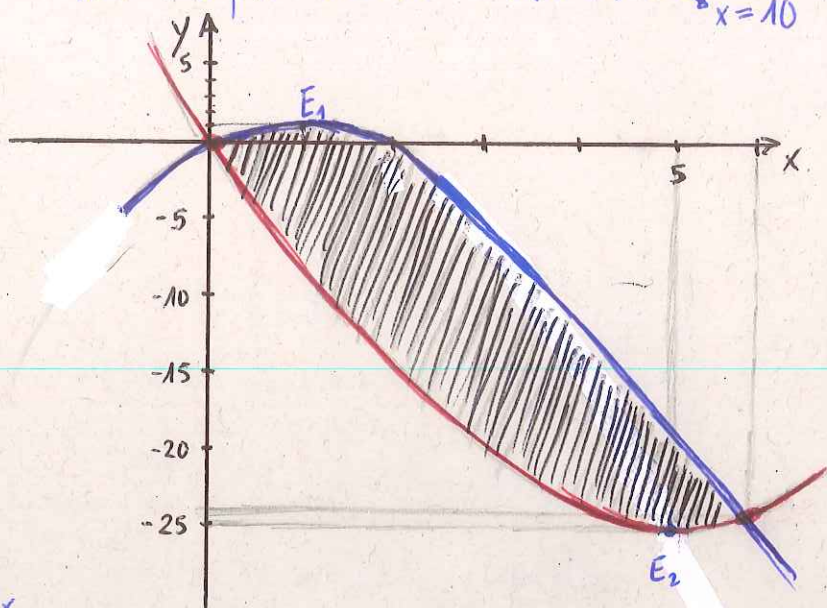
$$2x(-x + 6) = 0 \begin{cases} x=0 \rightarrow (0, 0) \\ x=6 \rightarrow (6, -24) \end{cases}$$

2. Kenketa-funtzioa

$$y = (x^2 - 10x) - (-x^2 + 2x) = 2x^2 - 12x$$

3. Integrala $\int_0^6 (2x^2 - 12x) dx = \left[\frac{2x^3}{3} - 6x \right]_0^6 = \left(\frac{2 \cdot 6^3}{3} - 36 \right) - (0 - 0) = 108$

4. Azalera $A = 108 u^2$



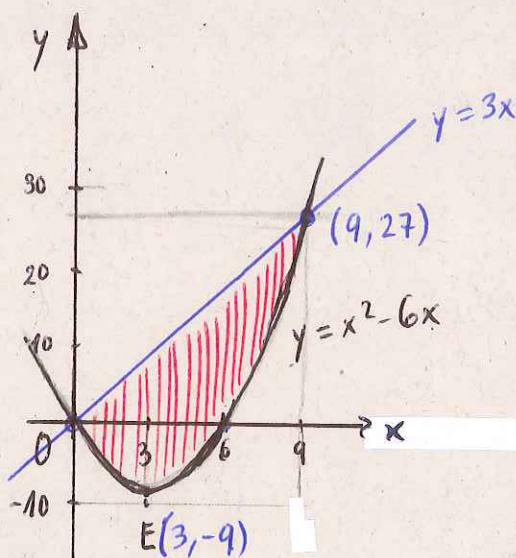
2014 UZTAILA A4

$$y = x^2 - 6x \rightarrow x = \frac{6}{2} = 3 \Rightarrow E(3, -9)$$

$$y = 3x \rightarrow (0, 0) \text{ eta } (3, 9) \text{ adibidez}$$

Parabolaren ebaki puntuak X ardatzarekin:

$$x^2 - 6x = 0 \begin{cases} x=0 \\ x=6 \end{cases}$$



1. Ebaki-puntuak funtzioen artean:

$$x^2 - 6x = 3x \rightarrow x^2 - 9x = 0 \begin{cases} x=0 \Rightarrow (0, 0) \\ x=9 \Rightarrow (9, 27) \end{cases}$$

2. Kenketa-funtzioa $y = 3x - (x^2 - 6x) = -x^2 + 9x$

3. Integrala

$$\int_0^9 (-x^2 + 9x) dx = \left[-\frac{x^3}{3} + \frac{9x^2}{2} \right]_0^9 = \left(-\frac{9^3}{3} + \frac{9 \cdot 9^2}{2} \right) - (0 + 0) = \frac{243}{2}$$

4. Azalera

$$A = \frac{243}{2} u^2$$

2013 EKAINA AY

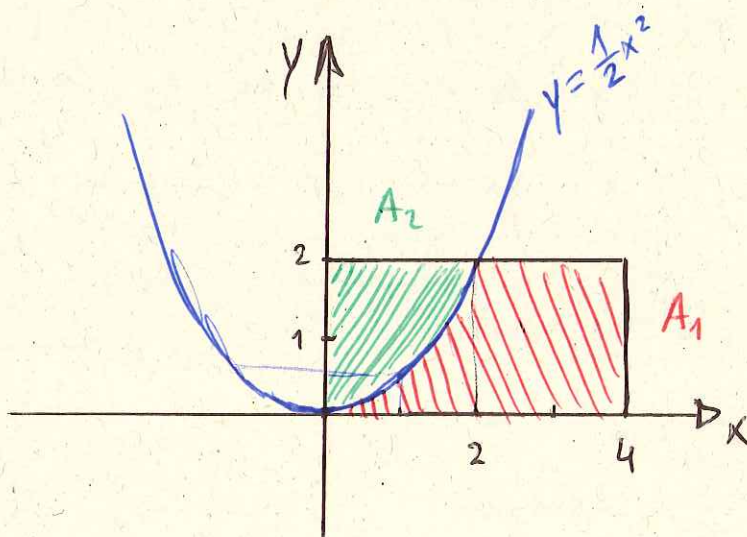
$$y = \frac{1}{2}x^2 \rightarrow \text{erpinu } E(0,0)$$

↳ Beste puntu batzuk $(1, \frac{1}{2})$

$(-1, \frac{1}{2})$

$(2, 2)$

$(2, -2)$



1. Ebaki puntuak $(0,0)$ eta $(2,2)$

2. Ez da beharrezkoa konketa funtzioa

3. Integralak □ konputua

$$(A_1) \rightarrow \int_0^2 \frac{1}{2}x^2 dx + 2 \cdot 2 = \left[\frac{x^3}{6} \right]_0^2 + 4 = \left(\frac{8}{6} - 0 \right) + 4 = \frac{16}{3}$$

4. Azalera:

$$\boxed{A_1 = \frac{16}{3} \text{ u}^2}$$

$$A_2 = \boxed{\text{shaded area}} - A_1 = 4 \cdot 2 - \frac{16}{3} = 8 - \frac{16}{3} \Rightarrow \boxed{A_2 = \frac{8}{3} \text{ u}^2}$$